

Pitch classes and timbre coordinates

FFT chroma, native ECQT, and the five MFCC coefficients.

03.01 · Folding octaves into chroma

EXPLANATION

Chroma combines energy from octave-related notes into twelve pitch classes. In sllr's ordinary descriptor pipeline, FFT bins between 27.5 and 4,186 Hz are assigned to the nearest equal-tempered note, then their magnitudes are summed and normalized.

This is a compact harmonic fingerprint, not a chord transcription. Harmonics, detuning, and coarse FFT spacing at low frequencies can distribute energy among several classes.

The structure view uses this FFT-derived chroma. The separate native CQT view has different windows and uses power when folding its pitch bins.

MATHS

For eligible positive-frequency bins,

$$q_k = \text{round}\left(69 + 12 \log_2 \frac{f_k}{440}\right).$$

The FFT chroma accumulator and normalization are

$$\tilde{C}[c] = \sum_{k: q_k \bmod 12 = c} A[k], \quad C[c] = \frac{\tilde{C}[c]}{\sum_d \tilde{C}[d]}.$$

A4 is 440 Hz and class zero is C. If the accumulator has negligible total magnitude, the code leaves a zero vector.

03.02 · Native constant-Q analysis

EXPLANATION

ECQT uses geometrically spaced frequencies and pitch-dependent window lengths. Low notes receive longer windows; high notes receive shorter ones. sllr's native wrapper exposes 84 pitch-bin magnitudes starting at C2.

Kernels are centered on the playback cursor, and the input is zero-padded at recording edges. Pitch analysis updates at approximately 10 Hz and reuses results between updates.

Long bass windows improve frequency discrimination at the cost of temporal precision. CQT pitch-class energy can be activated by harmonics; it is not a monophonic fundamental-frequency estimate.

MATHS

At twelve bins per octave, the nominal constant-Q parameters are

$$Q = \frac{1}{2^{1/12} - 1}, \quad f_k = 65.406391 \cdot 2^{k/12},$$

$$N_k = \left\lceil \frac{Q f_s}{f_k} \right\rceil, \quad f_s = 22050.$$

A conceptual pitch kernel measures a windowed complex sinusoidal projection at f_k ; ECQT evaluates sparse FFT-domain kernels. The native configuration uses Hamming windows and sparsification threshold 0.001. Folding magnitudes $|C_k|$ into pitch-class power gives

$$P[c] = \frac{\sum_{k \bmod 12=c} |C_k|^2}{\sum_k |C_k|^2}.$$

Silence yields zero instead of division by zero.

03.03 · MFCCs summarize the spectral envelope

EXPLANATION

MFCCs compress the broad shape of the spectrum through a mel-spaced filterbank, logarithms, and a cosine transform. slr uses 26 triangular filters and retains five coefficients, including coefficient zero.

These coefficients depend on the exact analysis conventions. The implementation has no pre-emphasis or liftering, and uses an unnormalized DCT-II. Coefficient zero includes level information.

Silence produces values determined by the logarithmic floor. Those values should not be interpreted as a measured timbre.

MATHS

Frequency is mapped to mel by

$$\text{mel}(f) = 2595 \log_{10}(1 + f/700).$$

For filter $B_b[k]$, define

$$E_b = \sum_k P[k] B_b[k], \quad \ell_b = \ln \max(E_b, 10^{-24}).$$

The retained coefficients are

$$c_j = \sum_{b=0}^{25} \ell_b \cos\left(\frac{\pi j(b + 1/2)}{26}\right), \quad j = 0, \dots, 4.$$

They are spectral-envelope coordinates rather than physical units or genre labels.